University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester 1 - UCSC AY21 [held in Sept./ Oct. 2024]

SCS 1307 — Probability and Statistics

(Two (2) Hours)

Answer ALL questions

Number of Pages = 08 Number of Questions = 04

To be completed by the candidate								
Index Number:								

94

Important Instructions to candidates:

- I. **For Question number 01 and 02, encircle the correct choice in the tables given in the Page 02.**
- II. **Students should answer to the question number 03 and 04 in the medium of English language, only in the answer booklets.**
- III. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- IV. Write your index number CLEARLY on each and every page of this question paper and answer booklets.
- V. Attach this question paper with answer booklets.
- VI. This paper consists of 04 questions in 09 pages (including the Cover Page).
- VII. Answer ALL questions.
- VIII. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- IX. Non-Programmable calculators are allowed
- X. Do not tear off any part of this question paper. Under no circumstances may this question paper, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Answers Table for Question 01

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice in the following table.

Question Number	Answers					Question Number	Answers				
i	a	b	c	d	e	ix	a	b	c	d	e
ii	a	b	c	d	e	x	a	b	c	d	e
iii	a	b	c	d	e	xi	a	b	c	d	e
iv	a	b	c	d	e	xii	a	b	c	d	e
v	a	b	c	d	e	xiii	a	b	c	d	e
vi	a	b	c	d	e	xiv	a	b	c	d	e
vii	a	b	c	d	e	xv	a	b	c	d	e
viii	a	b	c	d	e						

Answers Table for Question 02

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice in the following table.

Question Number	Answers					Question Number	Answers				
i	a	b	c	d	e	ix	a	b	c	d	e
ii	a	b	c	d	e	x	a	b	c	d	e
iii	a	b	c	d	e	xi	a	b	c	d	e
iv	a	b	c	d	e	xii	a	b	c	d	e
v	a	b	c	d	e	xiii	a	b	c	d	e
vi	a	b	c	d	e	xiv	a	b	c	d	e
vii	a	b	c	d	e	xv	a	b	c	d	e
viii	a	b	c	d	e						

Question 01 (25 Marks)

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice on the 2nd page of this question paper.

(i). Suppose that the weights of first year university students have symmetrical bell shaped distribution with the mean 60kg and the variance 49. Using the empirical rule in statistics, approximately what percentage of weights are between 60kg and 67kg?

a). 2.5% c). 34% e). 50%

b). 13.5% d). 47.5%

(ii). Let A, B and C be three events. Which of the following represents the event that all A, B and C occurs?

a). $A \cap B \cap C$ c). $A \cap B \cap C^c$ e). $A \cap B \cup C$

b). $A \cup B \cup C$ d). $A \cup B \cup C^c$

(iii). The standard deviation of a Poisson distribution with mean 16 is,

a). 3 c). 8 e). 256

b). 4 d). 16

(iv). Which of the following is an example of a discrete random variable?

a). The number of SMS received to your mobile phone in a particular day.

b). The amount of megabytes of a 1TB hard disk.

c). Height of a student in a volleyball team.

d). The number of players in a cricket team.

e). Mobile phone number of a student.

(v). If A and B are independent events, then always,

a). $P(A \cup B) = P(A) + P(B)$ d). $P(A \cup B) = P(A) - P(B)$

b). $P(A \cup B) = P(A) \times P(B)$ e). $P(A \cap B) = P(A) + P(B)$

c). $P(A \cap B) = P(A) \times P(B)$

(vi). The variance of a continuous uniform distribution which is defined in the range of (4,8) is,

a). $\frac{8+4}{2}$ c). $\frac{(4-8)^2}{2}$ e). $\frac{8-4}{2}$

b). $\frac{(8-4)^2}{12}$ d). $\frac{(8+4)^2}{12}$

(vii). When you randomly pick a card from an ordinary card pack, chance of getting a black card is,

a). 1/52 c). 1/26 e). 1/2

b). 1/4 d). 4/13

(viii). A pool of a cricket team contains five bowlers such that three are left handed and two are right handed bowlers. When two bowlers are selected randomly, the probability of having at least one right handed bowler is,

a). 12/25
b). 14/25
c). 14/20
d). 12/20
e). 4/25

(ix). X is a binomially distributed random variable with $P[X = 0] = 3.2 \times 10^{-4}$. Then the values of n and p are,

a). 5, 0.2
b). 4, 0.9
c). 5, 0.8
d). 16, 0.9
e). 2, 0.9

(x). Which of the following is always correct about continuous random variable with a probability density function $f(x)$?

a). $\int_{-\infty}^{+\infty} f(x)dx = 1$
b). $f(x) > 0$
c). $-\infty \leq f(x) \leq +\infty$
d). $\int_{0}^{+\infty} f(x)dx = 1$
e). $2\int_{0}^{+\infty} f(x)dx = 1$

(xi). Parameters of the binomially distributed random variable X are (with the standard notations),

a). x and n
b). x and p
c). n and p
d). np and np(1-p)
e). p and q

(xii). The variance of a Poissonly distributed random variable X, is 4. Then the mean of X is,

a). 4
b). 2
c). 16
d). 8
e). 1

(xiii). A uniformly distributed continuous random variable X is defined in the interval [-5, 3] , then $P[-3 < X < 1]$,is

a). 1/4
b). 1/2
c). 1/6
d). 1/8
e). 1/32

(xiv). Using the empirical rule in statistics, percentage of data within μ and $(\mu + 2\sigma)$ is,

a). 2.5%
b). 13.5%
c). 34%
d). 47.5%
e). 50%

(xv). The mean of a binomial distribution with n = 10 is 2. The standard deviation of this distribution is,

a). 0.2
b). 1.6
c). 0.4
d). 8.0
e). 1.26

Question 02 (25 Marks)

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice on the 2^{nd} page of this question paper.

(i). Let A and B be two events. Which of the following represents the event that none of the two events occurs?

a). $(A \cap B)^C$ c). $A^C \cap B$ e). $A \cap B$

b). $(A \cup B)^C$ d). $A \cup B^C$

(ii). If $P(A) = 0.4$, $P(B^c) = 0.4$ and $P(B \cup A) = 0.8$ then $P(B \mid A)$ is,

a). 0.26 b). 0.40 c). 0.50 d). 0.60 e). 0.80

(iii). If $P(A) = 0.4$, $P(B \mid A) = 0.3$ and $P(A \mid B) = 0.4$ then $P(B^c)$ is,

a). 0.06 b). 0.26 c). 0.30 d). 0.60 e). 0.70

(iv). If $P(A|B) = 0.6$, $P(B) = 0.4$ and $P(A \cup B) = 0.6$, then $P(A)$ is:

a). 0.14 b). 0.24 c). 0.34 d). 0.44 e). 0.54

(v). If events A and B are mutually exclusive, then always,

a). $P(A \cup B) = P(A) + P(B)$

b). $P(A \cup B) = P(A) \times P(B)$

c). $P(A \cap B) = P(A) \times P(B)$

d). $P(A \cup B) = P(A) - P(B)$

e). $P(A \cap B) = P(A) + P(B)$

(vi). Let $Z \sim N(0,1)$ and $P(Z < a) = 0.9750$, then the value of a is,

a). -1.96 b). -1.64 c). 0 d). 1.64 e). 1.96

(vii). Let $Z \sim N(0,1)$ and $P(|Z| < a) = 0.9500$, then the value of $|a|$ is,

a). 1.20 b). 1.26 c). 1.44 d). 1.60 e). 1.96

(viii). A box contains four red balls and five blue balls. When two balls are selected at random with replacement, the probability of obtaining exactly one red ball is;

a). 32/72 b). 25/72 c). 52/72 d). 20/81 e). 50/81

(ix). The probability of a certain event is,

a). 1 b). -1 c). 0.1 d). 0 e). 0.5

(x). Suppose that 10% trains arrive on time to a certain railway station. Let X be the number of trains arrived on time in a sample of size 300. The exact distribution of X is;

a). Binomial with n = 300 and p = 0.1

b). Binomial with n = 300 and p = 10

c). Binomial with n = 300 and p = 0.01

d). Poisson with $\lambda = 0.1$

e). Poisson with $\lambda = 30$

(xi). If $P(A|B) = 0$ then;

a). A and B are impossible events.

b). A and B are mutually exclusive.

c). $P(A \cap B) = P(A)P(B)$

d). $A \subset B$

e). $B \subset A$

(xii). A committee of two executives is randomly selected from a group of five male and four female executives. The probability that the committee contains exactly two female executive is,

a). $\frac{{}^5C_0\ {}^4C_2}{{}^9C_2}$

b). $1 - \frac{{}^5C_0\ {}^4C_2}{{}^9C_2}$

c). $\frac{{}^5C_0\ {}^4C_2}{{}^9C_2} - 1$

d). $\frac{{}^5C_2\ {}^4C_0}{{}^9C_2}$

e). 2/9

(xiii). If the standard deviation of a Poisson distribution is 3 then the mean is:

a). 1.50 b). 1.73 c). 3.00 d). 6.00 e). 9.00

(A similar question is in Question 1. The same knowledge is evaluated)

(xiv). Let X be a random variable with V[X] = 4. Then the standard deviation of $Y = \frac{1}{4}X + 1$ is,

a). 2 b). 1.5 c). 5 d). 6 e). 8

(xv). Which one of the following does not represent a valid probability distribution function?

a). $P[X = x] = (x + 2)/25, \quad x = 1, 2, 3, 4, 5$

b). $P[X = x] = (x + 2)/15, \quad x = -1, 0, 1, 2, 3$

c). $P[X = x] = (x - 2)/5, \quad x = 1, 2, 3, 4, 5$

d). $P[X = x] = 1/5, \quad x = -2, -1, 0, 1, 2$

e). $P[X = x] = x/15, \quad x = 1, 2, 3, 4, 5$

Question 03 (25 Marks)

a) State the classical definition of probability. (04 Marks)

b) If A and B are two events on a sample space S, and $B \subset A$ with the usual probability notations, prove that $P(B) < P(A)$. (05 Marks)

c) Kushani decided to improve her Mathematics and Statistics knowledge by attending tuition classes. The probability that Kushani enrolls a Mathematics class is 0.60 and the probability that she enrolls a Science class is 0.30. The probability that she enrolls a Mathematics class given that she enrolls a Science class is 0.25. (08 Marks)

   i). What is the probability that Kushani enrolls in both Mathematics and Science classes?
   ii). What is the probability that Kushani enrolls in Mathematics or Science?
   iii). Are enrolling in Mathematics and Science classes independent? Justify your answer.
   iv). Are enrolling in Mathematics and Science mutually exclusive events?

d) It is estimated that 60% of emails are spam emails. A software has been applied to filter these spam emails before they reach to inbox. A certain brand of software claims that it can detect 90% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 10%. If an email is detected as spam, then what is the probability that it is in fact a spam email? (08 Marks)

Question 04 (25 Marks)

a) The probability of a student having stomach-ache during the examination is 0.01. A sample of 10 students were taken randomly. (05 Marks)

   i). Calculate the probability that exactly one student will have stomach-ache during the examination.
   ii). Calculate the standard deviation of the number of students will have stomach-ache during the examination.

b) Nayani is a first year student of UCSC and she receives on average λ number of calls in an hour. Let X be the random variable, number of calls receive in an hour. (08 Marks)

   i). Is Poisson distribution suitable to model the random variable X. Justify your answer.
   ii). Assuming $X \sim \text{Poisson}(\lambda)$, if $2P(X=0) = P(X=2)$, show that $\lambda = 2$.
   iii). Calculate the probability that Nayani will receive at least 2 calls in an hour.
   iv). Calculate the probability that Nayani will receive only one call in four hours.

c) Time taken to complete a particular type of task is normally distributed, with average 40 minutes and standard deviation 4 minutes. If 90 such tasks are considered (12 Marks)

   i). Calculate the expected number of tasks that take less than 45 minutes.
   ii). Calculate the expected number of tasks that take time between 30 minutes and 47 minutes
   iii). Determine the cut-off time points for the central 90% times take to complete.

***************** The End of the Question Paper *****************

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997